Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/14/2015
PI Name:	Poirier, David Sc.D.		
Project Title:	Effect of Varying Convection on Dendrite Morphology and Macrosegregation		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	poirierd@email.arizona.edu	**Fax:**	FY 520-621-8059
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-6072
Organization Name:	The University of Arizona		
PI Address 1:	1235 East James E Rogers Way		
PI Address 2:	Department of Materials Science and Engineering		
PI Web Page:	http://www.mse.arizona.edu/		
City:	Tucson	**State:**	AZ
Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2010	**End Date:**	11/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2014 per NSSC information (Ed., 9/2/13)		
Key Personnel Changes/Previous PI:	Professor Jhon Jairo Olaya Florez of the Universidad Nacional de Colombia, Bogata spent part of his sabbatical with the research team at The University of Arizona during February-August of 2013.		
COI Name (Institution):	Erdmann, Robert (University of Arizona) Tewari, Surendra (Cleveland State University)		
Grant/Contract No.:	NNX10AV40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research is ground-based research on directional solidification (DS) of metallic alloys. The investigators are David Poirier (DP) and Robert Erdmann of The University of Arizona and Surendra Tewari (ST) of Cleveland State University. DP and ST have collaborated for many years, and the requisite facilities and team are in place. Hence, the research is cost-effective and promises to achieve "flight status" in minimal time. The research of this grant pertains to convection that is driven by DS through changes in cross-sections (because of solidification shrinkage) and thermosolutal convection. In microgravity, of course, the thermosolutal convection will be mitigated so that effects of the shrinkage-driven flow in cross-sectional changes can be isolated from the usual thermosolutal convection that always occurs when solidification occurs on Earth. The outcome of the research is expected to be relevant to commercial DS-castings, that are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and electric power generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research not only has scientific merit, it is also relevant to the production of commercial DS castings. The proposed ground-based research comprises the following: 1) directional solidification experiments that are compatible with the solidification cartridge assemblies (SCAs) that can be employed in a furnace on the International Space Station (ISS). The research comprises: 1) selected alloys have varying density-responses to gravity forces; 2) crucible designs with variable cross-sections but compatible with the current SCAs; 3) sophisticated characterizations of the solidified microstructures, including both micro- and macro-segregation; and 4) supporting computer simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The outcome of the research is expected to be relevant to commercial directionally solidified-castings, that are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and turbine electric power-generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research is very much needed for its scientific merit and also because it is relevant to the production of commercial DS castings.
Task Progress:	Directional solidification experiments were done at Cleveland State University. Specimens from the casting-ingots that were directionally solidified with changes in cross-section were taken in order to examine the cast microstructure and the macrosegreagtion patterns. Macrosegregation was measured and compared to macrosegregation predicted by computer simulations. Good agreement was found between the simulated results and the empirical results.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Poirier DR. "Density, viscosity, and diffusion coefficients in hypoeutectic Al-Si liquid alloys: An assessment of available data." Metallurgical and Materials Transactions B. 2014 Aug;45(4):1345-54. http://dx.doi.org/10.1007/s11663-014-0037-8 , Aug-2014
Papers from Meeting Proceedings	Lauer MA, Poirier DR, Erdmann RG, Johnson L, Tewari SN. "Simulations of the Effects of Mold Properties on Directional Solidification." Presented at the International Mechanical Engineering Conference and Exposition, San Diego, California, November 15-21, 2013. ASME 2013 International Mechanical Engineering Congress and Exposition. Volume 2B: Advanced Manufacturing. p. 56192-56203. Paper No. IMECE2013-66830. http://dx.doi.org/10.1115/IMECE2013-66830 , Dec-2013